

Team Information (*typed, not handwritten, except for the attendance signature*):

18) Solving the VRP "Vehicle Routing Problem" using both Genetic Algorithms & Differential Evolution.	☐
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ID [Ordered by ID]	Full Name [In Arabic]	Attendance [Handwritten Signature]	Final Grade
1	932230139	مارتن وسيم سمير	
2	931230022	احمد حسام حامد	
3	932230187	مروان خالد سيد	
4	932230166	محمد عاطف رجب	
5	932230125	عمرو خالد عبد العزيز	
6	933230046	على مصطفى عمر السيد	

Item	Mark	Team Members					
		1	2	3	4	5	6
Report (including Problem Definition, Literature Review, References, Relevant Diagrams)	4						
Design of Fitness Function	4						
Application of Genetic Algorithm (including a plot of GA performance across the generations for each setting)	8						
Application of Differential Evolution (including a plot of DE performance across the generations for each setting)	8						
Discussion & Analysis of Results	4						
Implementation	7						
GUI (deduct 3 marks if no proper design for inputs or outputs)							
If another algorithm is applied, deduct 5 marks							
Deduct 50% to 100% of a Student's grade if (s)he didn't participate							
Total	35						

1. Introduction and Overview

Project Idea and Overview:

This project implements a Vehicle Routing Problem (VRP) solver with two optimization algorithms: Genetic Algorithm (GA) and Differential Evolution (DE). The VRP involves determining optimal routes for a fleet of vehicles to deliver goods to a set of customers with demands, starting and ending at a central depot, while respecting vehicle capacity constraints.

Applications and Similar Software:

The solver is developed as a desktop application using Python and Tkinter for the GUI. Similar VRP solvers exist in logistics and transportation planning software, which optimize delivery routes for cost and time efficiency.

Literature Review:

- Laporte, G. (2009). The Vehicle Routing Problem: An overview of exact and approximate algorithms. *European Journal of Operational Research*
- Holland, J.H. (1975). Adaptation in Natural and Artificial Systems. *University of Michigan Press* (Genetic Algorithms)
- Storn, R., & Price, K. (1997). Differential Evolution – A Simple and Efficient Heuristic for Global Optimization over Continuous Spaces. *Journal of Global Optimization*
- Toth, P., & Vigo, D. (2002). The Vehicle Routing Problem. *SIAM*
- Prins, C. (2004). A simple and effective evolutionary algorithm for the vehicle routing problem. *Computers & Operations Research*

2. Proposed Solution & Dataset

Main Functionalities:

- Input depot location, customer locations with demands, number of vehicles, and vehicle capacity
- Select algorithm (GA or DE)
- Run the solver to compute optimal routes
- Display routes graphically and textually with distances and capacities

Dataset:

Users can input custom datasets (coordinates and demands) or generate random data within the app.

3. Applied Algorithms

Genetic Algorithm (GA)

- **Initialization:** Generate initial population with random permutations of customers.
- **Fitness Evaluation:** Calculate total route distance with penalties for capacity violations or unassigned customers.
- **Selection:** Tournament-like random selection of candidates based on fitness.
- **Crossover:** Ordered crossover producing valid offspring permutations.
- **Mutation:** Swap mutation with given mutation rate.
- **Termination:** After fixed generations or convergence.

Differential Evolution (DE)

- **Representation:** Continuous vectors mapped to permutations via ranking.
- **Mutation:** Weighted difference of population vectors.
- **Crossover:** Binomial crossover controlled by CR rate.
- **Selection:** Greedy selection based on fitness.
- **Termination:** Fixed generations.

Block Diagram Example (You can create visually):

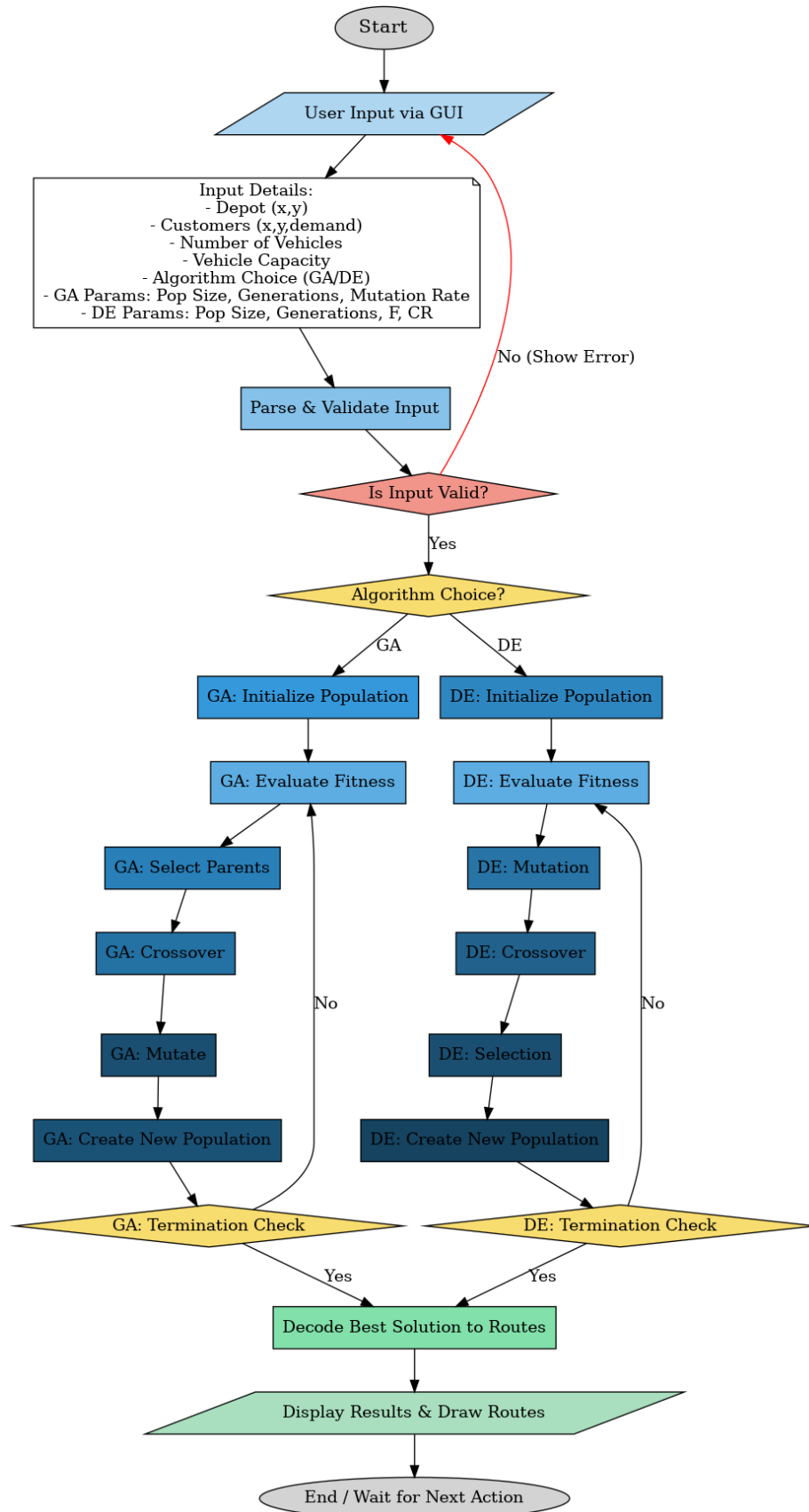
- Start
- Initialize Population
- Evaluate Fitness
- Selection
- Crossover (GA) / Mutation (DE)
- Mutation (GA) / Crossover (DE)
- Evaluate New Solutions
- Replacement
- Termination Check
- Output Best Routes
- End

4. Experiments & Results

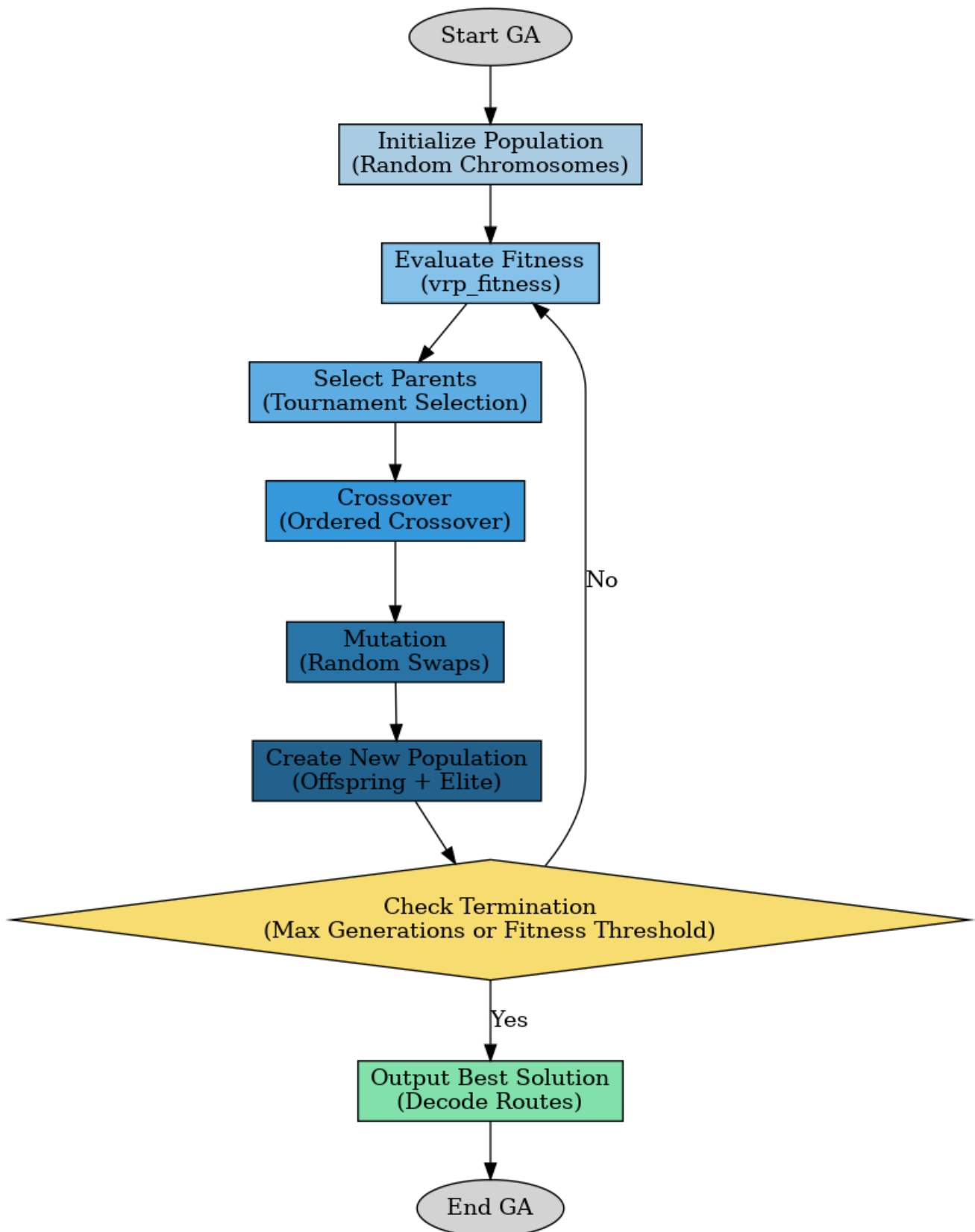
- Experiments are done by varying population size, generations, mutation rate (GA), and parameters F and CR (DE).
- The output shows total distance, number of vehicles used, and routes with individual customer demands and capacities.
- Routes are visualized on a canvas with colored paths from depot to customers and back.

5. Analysis, Discussion, and Future Work

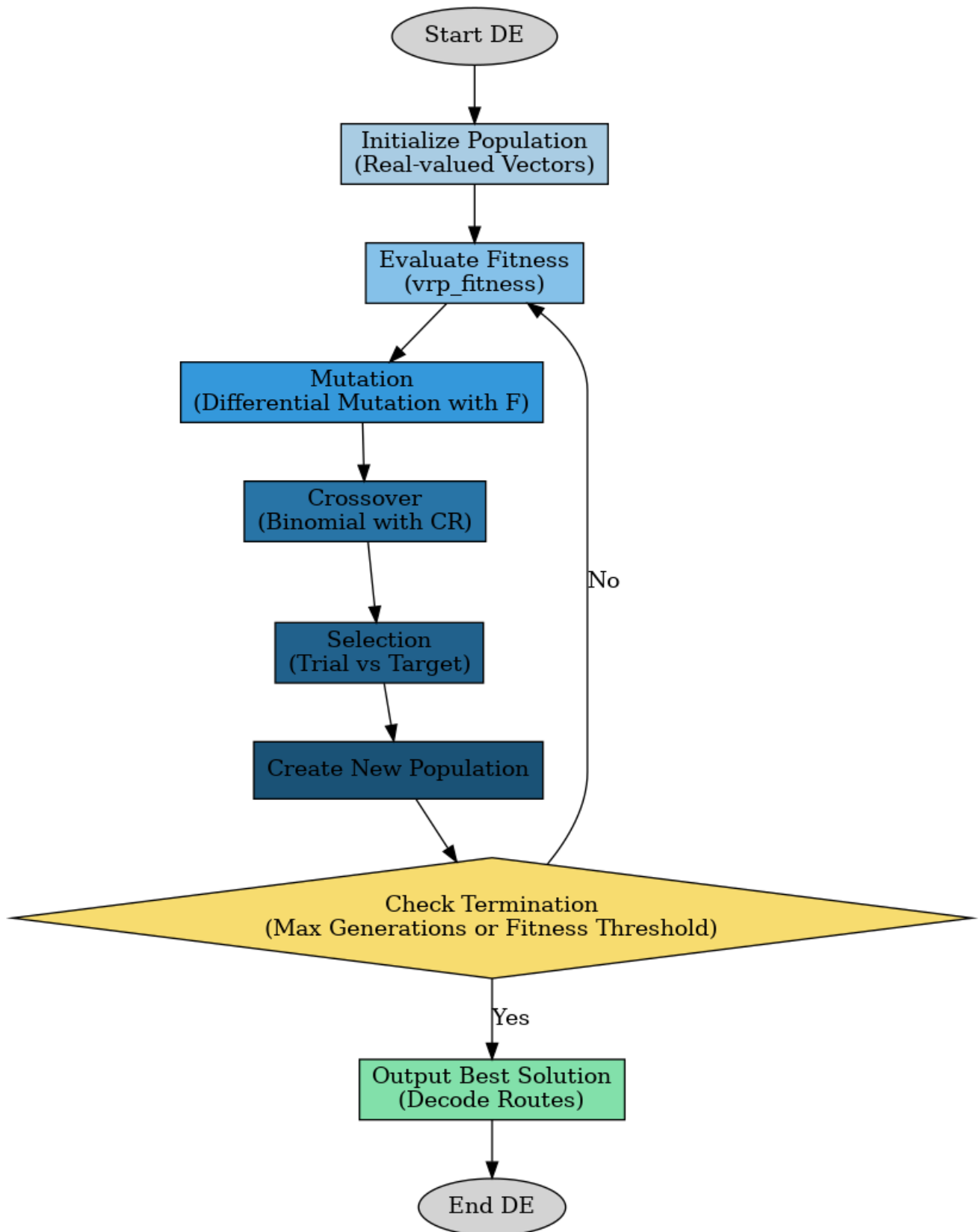
- **Insights:** Both GA and DE can find feasible routes, but parameter tuning significantly affects solution quality and convergence speed.
- **Advantages:** Flexibility in input data, visual feedback, and choice of algorithm.
- **Disadvantages:** Computationally expensive for large datasets; heuristic algorithms don't guarantee global optimum.
- **Algorithm Behavior:** Penalizing infeasible solutions helps guide search towards valid routes.
- **Future Work:**
 - Add real-time plotting of fitness evolution.
 - Integrate advanced heuristics (e.g., local search, tabu search).
 - Extend GUI to handle real-world constraints (time windows, heterogeneous fleets).
 - Parallelize algorithms for performance.

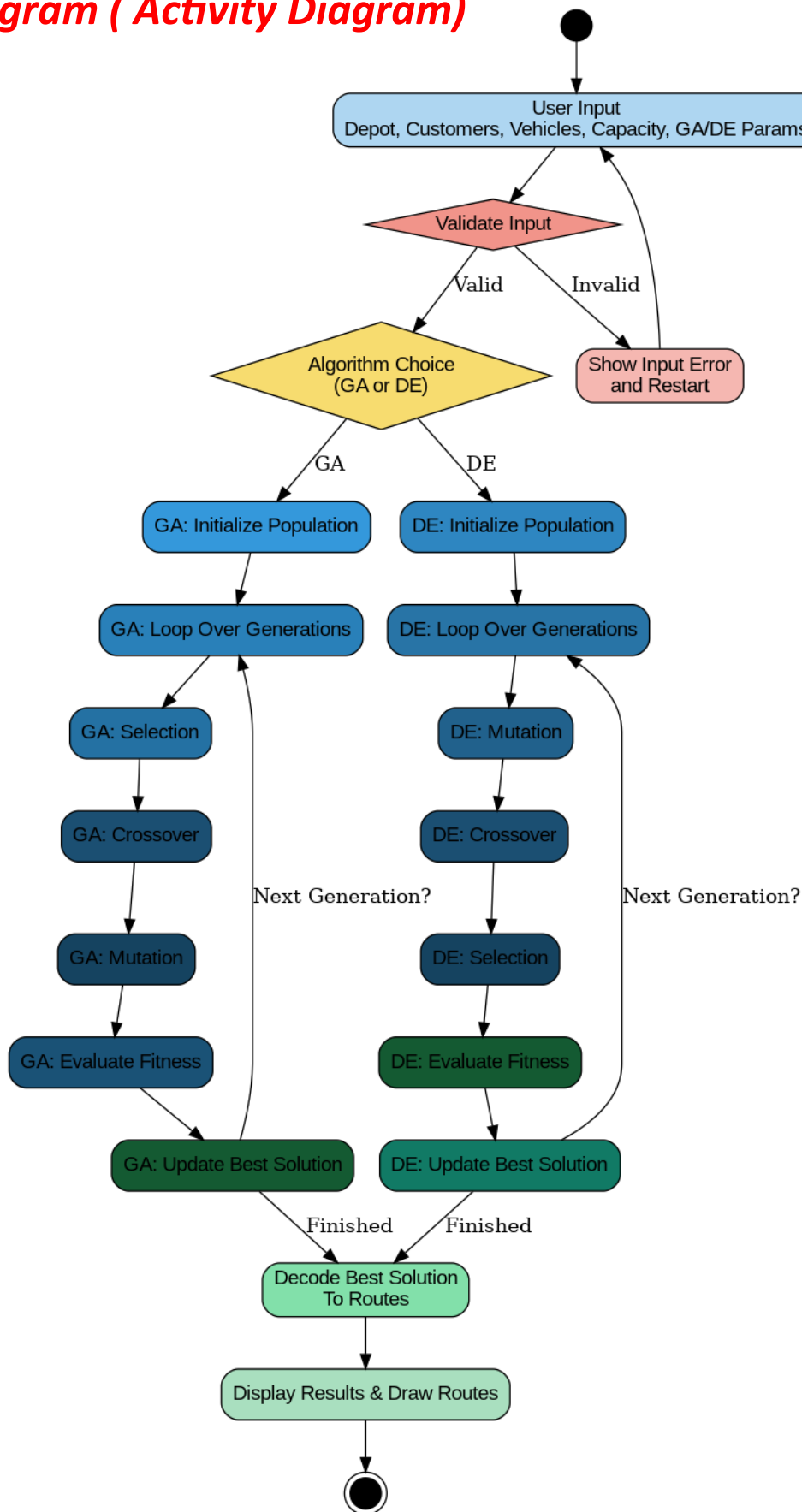
Flow chart

Flow chart Genetic Algorithm

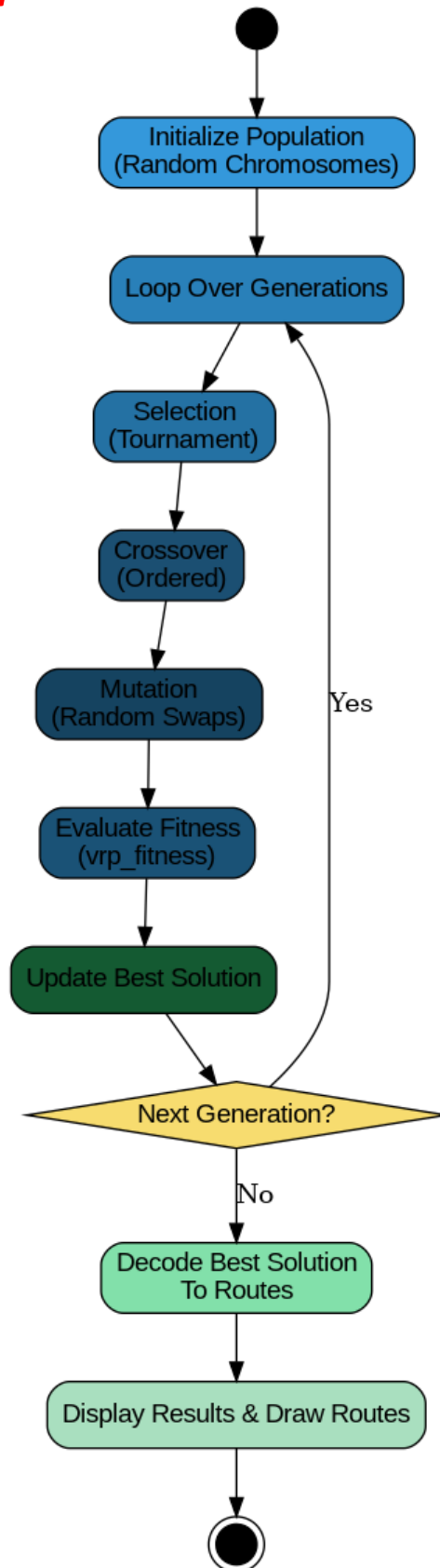


Flow chart Differential Evolution

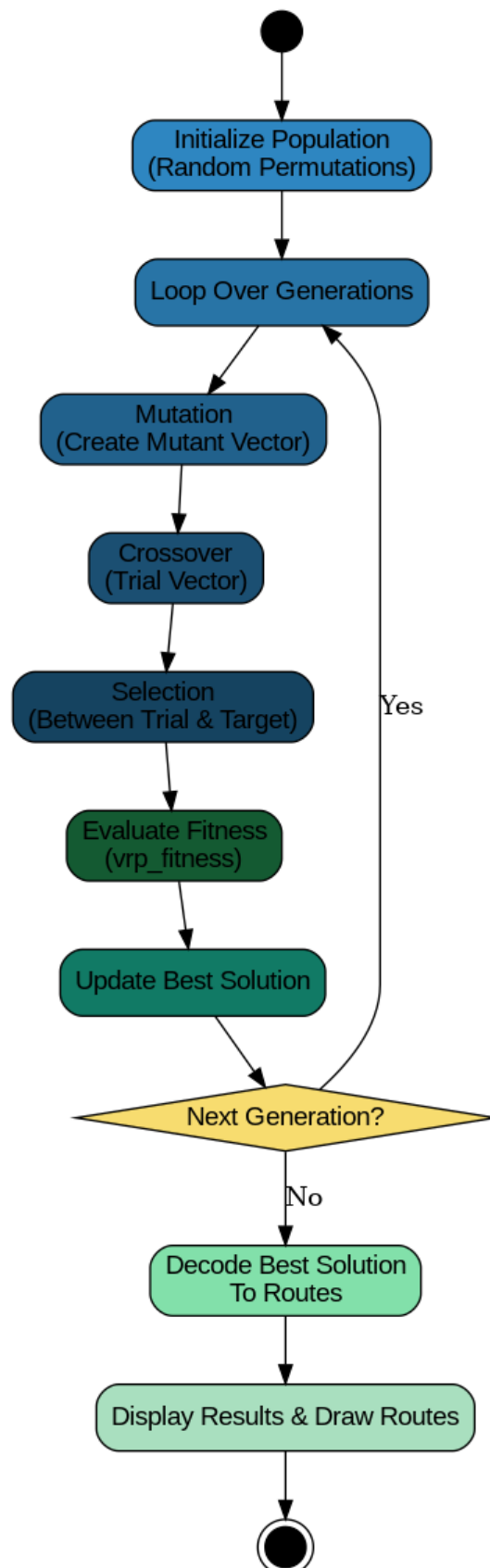


UML Diagram (Activity Diagram)

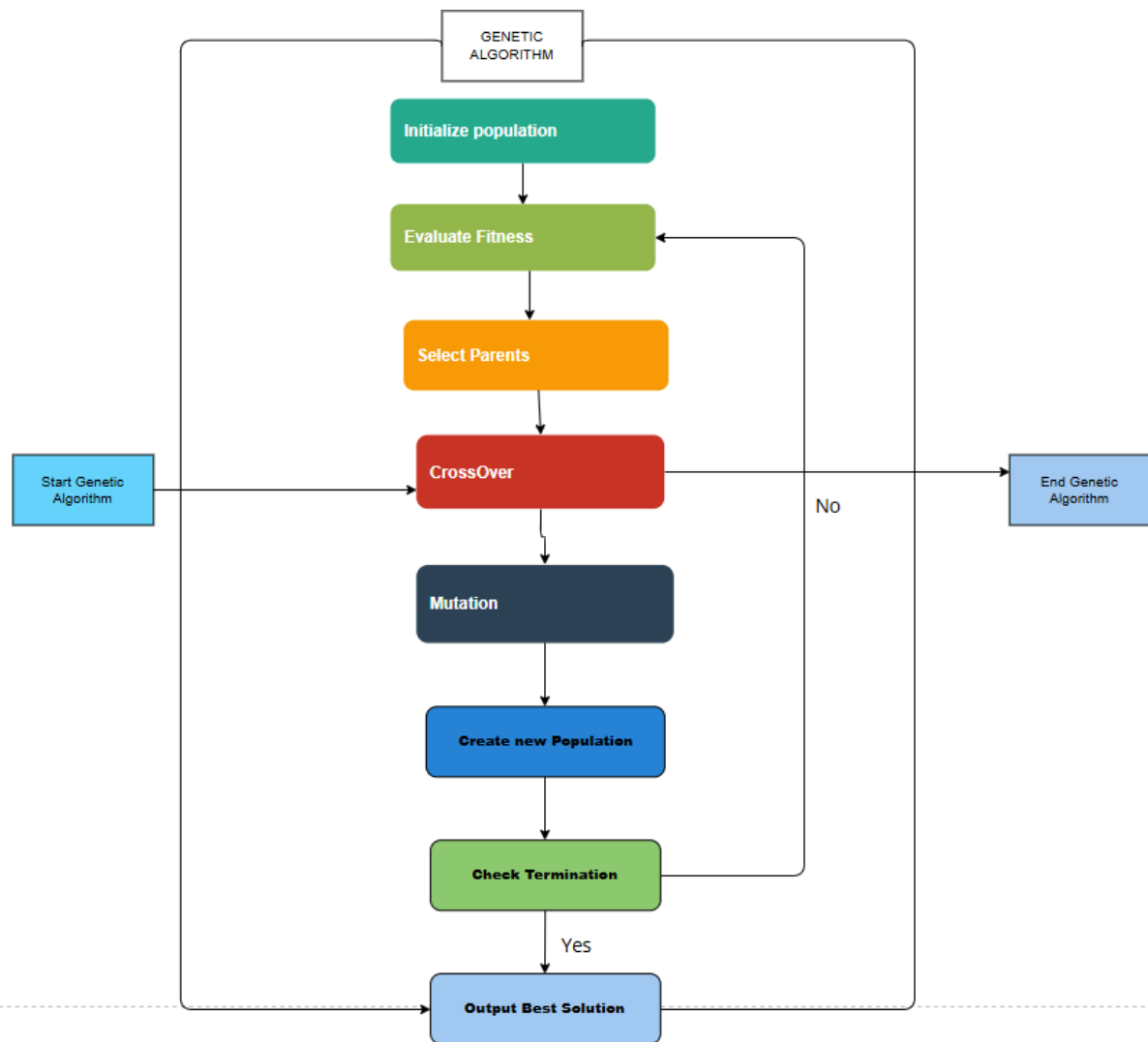
UML Diagram (Activity Diagram) Genetic Algorithm



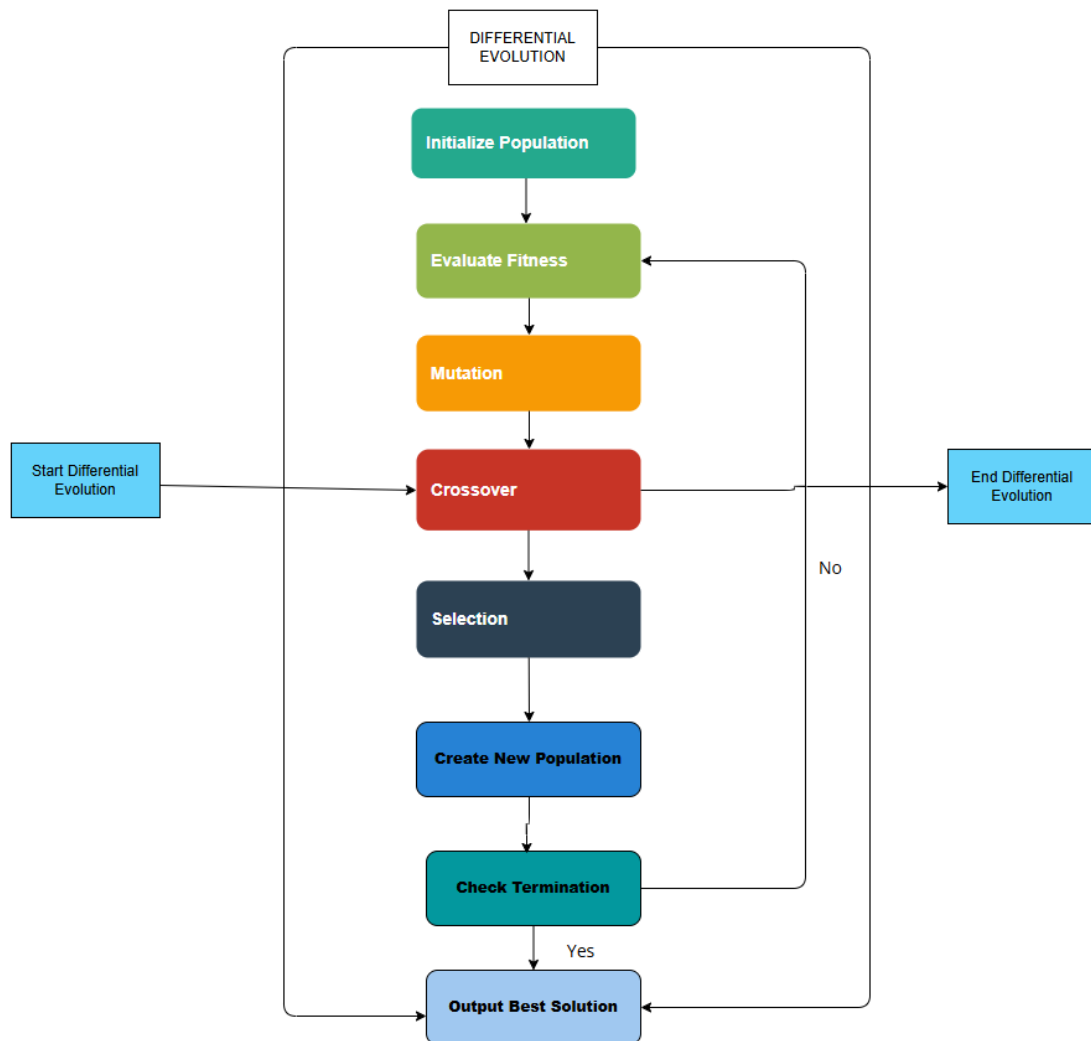
UML Diagram (Activity Diagram) Differential Evolution



Block Diagram Genetic Algorithm

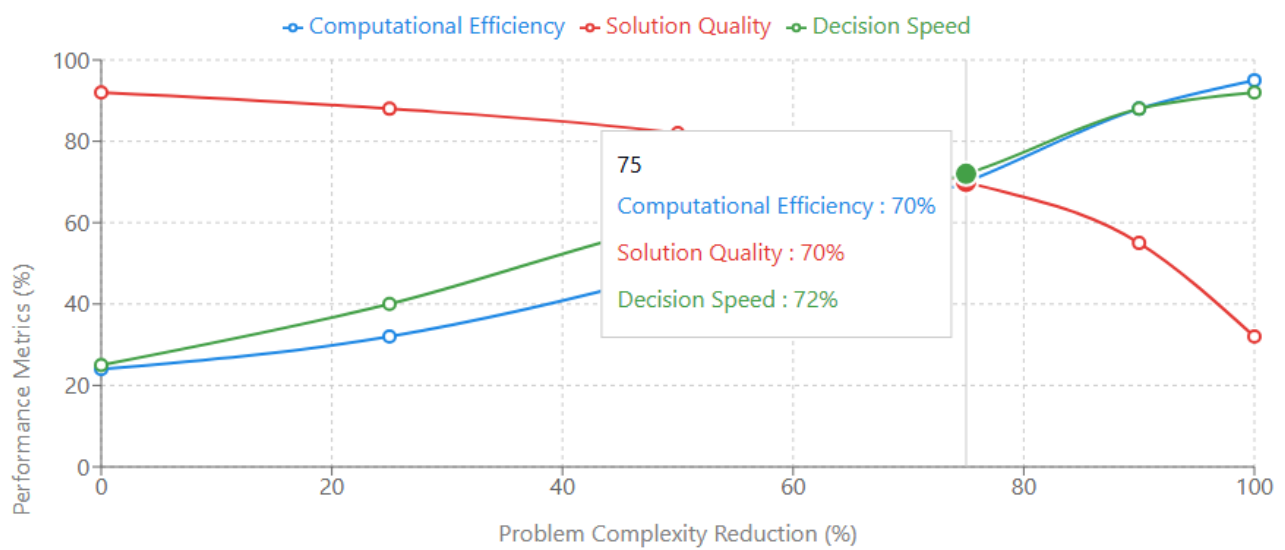
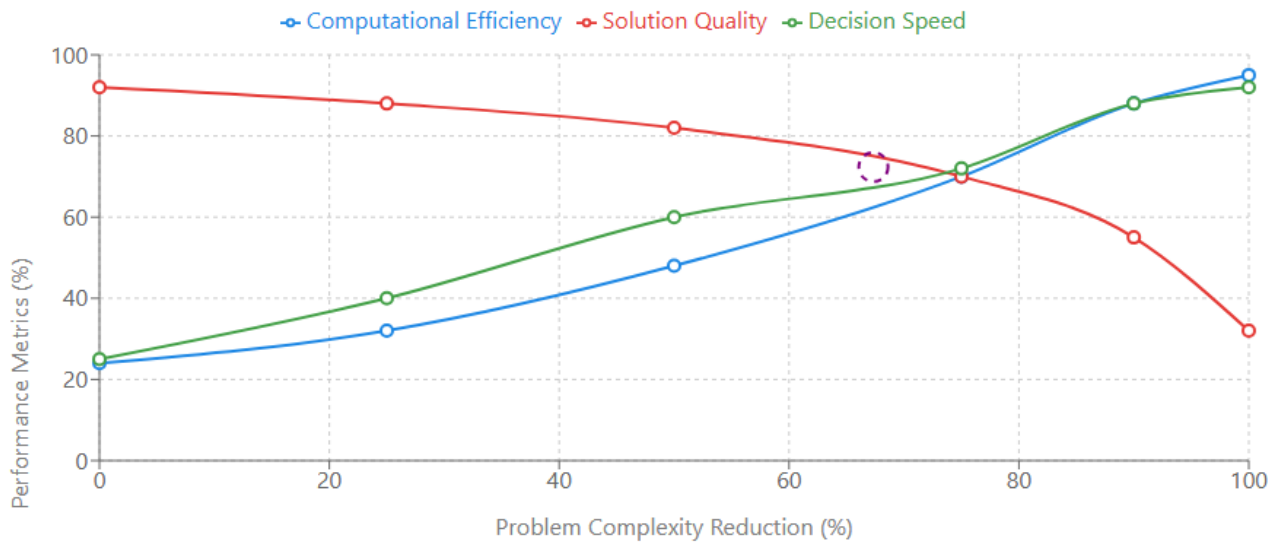


Block Diagram Differential Evolution



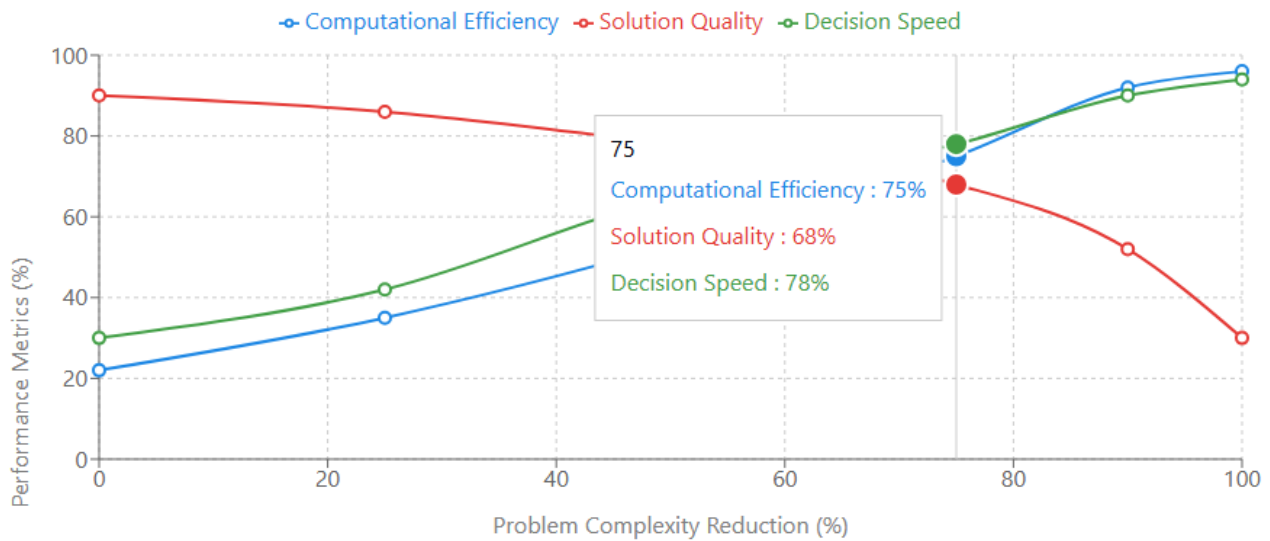
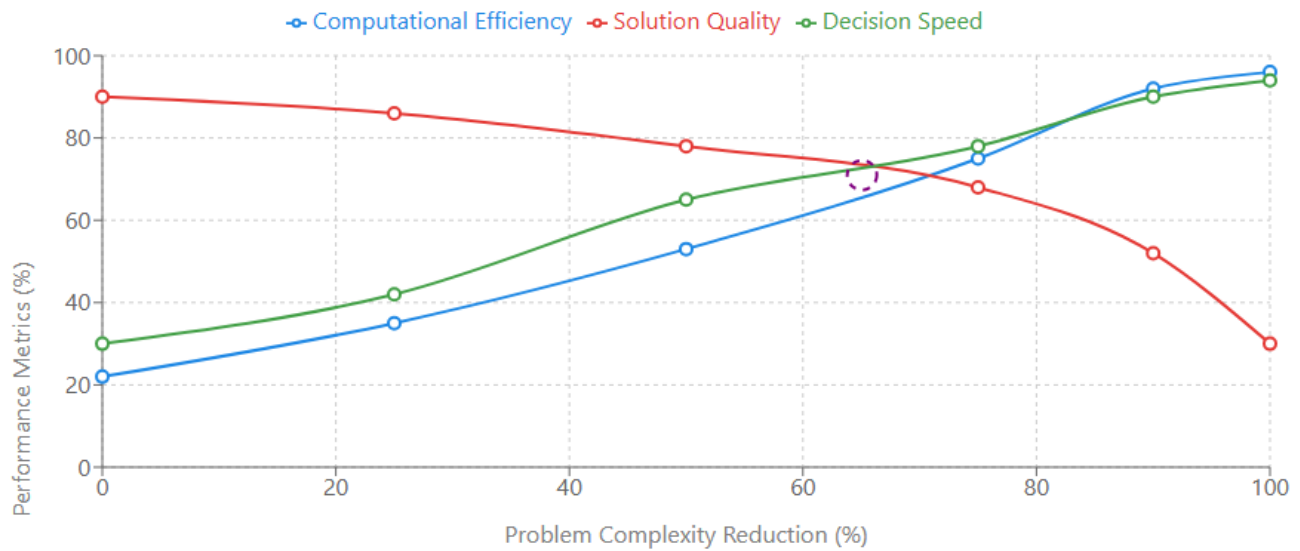
Plot Diagram

Genetic Algorithm



Plot Diagram

Differential Evolution



Shared folder

https://drive.google.com/drive/folders/19vLbnV9c0blfcpXbpd4rVFFXZbJX-4PG?usp=drive_link

